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Project Title: AirAware

1.0 Introduction

1.1 Background of Case Study

Due to severe air pollution caused by vehicle exhaust emissions, industrial activities, open burning, and forest fires, air quality in many cities, communities, and rural areas has been continuously deteriorating. In Southeast Asia, including Indonesia, people remain unaware of the risks caused by long-term exposure to harmful air and lack knowledge on how to protect themselves.

Even though Air Quality Index (AQI) and PM2.5 data are available online, many students, teachers, and families still find the information too technical and hard to understand. As a result, they are unsure how pollution levels affect their daily activities.

To address these issues, we propose AirAware AI – Personal Air Pollution Guide, an AI-powered website that turns complex air quality data into simple, clear, and practical guidance. The platform delivers personalized safety recommendations through air pollution education, real-time alerts, and localized pollution analysis. By combining environmental data with AI, this project makes air quality information easier to understand and helps people make safer everyday decisions.

1.2 Chosen SDG: Identify the specific Sustainable Development Goal

1. SDG 3 - Good Health and Well-being

AirAware AI helps users gain a deeper understanding of the potential risks of air pollution and methods for self-protection. The platform provides personalized health advice to help people minimize pollution impacts and cultivate healthier lifestyle habits. For example, it advises limiting outdoor activities when the Air Quality Index (AQI) is high.

2. SDG 4 - Quality Education

This platform acts as an educational tool for urban residents,outdoor activities users and drivers using electric cars. It provides interactive explanations and AI-powered Q&A functionality. This helps them understand the factors affecting air pollution and its impact on personal health. Not only does this enhance environmental awareness, but it also indirectly helps people develop the ability to make informed decisions.

3. SDG 13 - Climate Action

AirAware AI enhances environmental responsibility by analyzing local pollution sources such as toxic gasses from vehicle emissions and dust from construction sites to notify the health impact level to users. It simultaneously proposes specific solutions to reduce individual polluting behaviors. This project is primarily dedicated to helping people change their daily habits, thereby enabling climate action and fostering sustainable lifestyles.

1.3 Problem Statement

1. Lack of air pollution awareness in urban areas
- It is because many urban residents still do not understand how air pollution can affect their health.
2. Unsafe outdoor exercise behavior during polluted condition
- Some individuals are still jogging or cycling in polluted environments and inhaling the toxic gases emitted by vehicles or dust from construction sites.
3. The existing platform mainly focuses on displaying the AQI data monitoring on the interactive map.
- The existing not provide further explanation about the topic of air pollution and the recommendation to inform users the action should be taken during air pollution.
4. A lot of residents prefer to use a private car that is fuel-based due to its convenience and comfort.
- Rapidly increasing the number of vehicles causing emission of toxic gases becomes higher which means the surrounding temperature becomes higher. Many people acknowledge that cars are a major source of pollution but still prefer driving over public transport due to issues with efficiency or comfort.

<u>Key Features:</u>

A.) Real-time air quality levels display on gauge and interactive map

- Show current AQI levels for different urban areas in Malaysia through a colour-coded map system

- Green (AQI 0 - 50) - Good
- Yellow (AQI 51 - 100) - Moderate
- Orange (AQI 101 - 150)- Unhealthy for Sensitive Groups
- Red (AQI 151 - 200)- Unhealthy
- Purple (AQI 201 - 300)- Very Unhealthy
- Dark purple (AQI 301 - 500) - Hazardous

B.) AI Chatbot

Users can ask questions about the topic of air pollution to enhance their understanding such as air quality status, health risks, pollution sources and recommended precautions.

C.) Personal Health Advisor

-AI assistants can give several suggestions such as respiratory protection tips, outdoor advisory and ways to have a proper diet during air pollution based on the information from health profiles filled by users.

D.) Air Insights

-Provide latest global and local news about the air pollution.

-Receiving the announcement of upcoming environmental events**.** In addition, they can gain the benefits from the event by referring to the event highlight. For example, car free day is a jogging activity and provides the opportunity for users to have an electric car test drive session in order to enlighten them about the benefits of using electric cars.

-Informing the user about the factors that could cause the air pollution such as traffic congestion can significantly increase PM2.5 levels in urban areas, posing severe risks to commuters.

-Providing tips and guidelines to protect personal health during the period of poor air quality such as choosing appropriate masks or limiting outdoor activities.

E.) Smart AI assistant

Smart Route Suggestion:

Users can select the activity because the AI assistant can suggest the route or area that is less polluted for the user.

Smart Driving Health:

These features are designed to integrate into electric cars (EVs). The driver can view the reduction of carbon dioxide (CO_2), cabin air quality, primary pollutants, traffic exposure and smoke detection etc.

Target user:

- Urban residents
- Outdoor activities users
- Driver using electricity car

1.4 Objectives

1.4.1 Increase public awareness

- Aim to improve the understanding of real-time air pollution conditions among urban residents in Malaysia through simple and easy-to-understand data presentation (such as realistic images or similar metaphors)

1.4.2 Promote safer outdoor activities

- AI-powered intelligent recommendation systems will combine Air Quality Index, AQI with the day's weather data to help users choose safer, healthier, and more suitable locations for jogging or outdoor activities.

1.4.3 Provide personalized health guidance

- AI chatbots will provide users with more personalized Q&A, including but not limited to health advice and protective guidance related to air pollution.

1.4.4 Reduce health risk

- Reduce the dual risks to respiratory health and personal well-being that residents face from prolonged exposure to polluted environments due to a lack of awareness of air quality.

1.4.5 Support SDG 3 & 11

- Support Sustainable Development Goal 3 - **Good Health and Well-being** and Sustainable Development Goal 11 - **Sustainable Cities and Communities**.

2.0 Review of Existing Solutions

2.1 Analyze current solutions or competitors in the market

Solution	Provider	Main Function	Strengths	Limitations
World Air Quality Index (WAQI)	Global platform	Real-time global AQI monitoring and color-coded maps	- Real-time data - Global coverage - Easy visualization	- No personalized recommendation - No AI assistant - Not localized for Malaysia
Malaysia Air Pollutant Index (API)	Department of Environment Malaysia	Official API readings by state	- Official government data - Reliable source	- Basic interface - No health guidance - No activity recommendation
IQAir / AirVisual	IQAir (Switzerland)	AQI monitoring + air quality forecast + health suggestions	- Mobile app available - Forecast feature - Health suggestion	- Some features require payment - No exercise location recommendation - No conversational AI

2.2 Highlight gaps or limitations that AirAware aims to fix

2.2.1 Lack of Personalized Guidance

- Most existing air quality platforms only provide raw data such as Air Quality Index (AQI) values and pollution levels. However, they cannot offer more personalized advice based on the user's current health condition or daily activities.
- AirAware addresses this limitation by integrating an AI-powered personal health advisor and chatbot that can provide customized advice based on the user's current health condition and the currently detected air quality.

2.2.2 Limited Support for Safe Outdoor Activity Planning

- Current solutions primarily focus on displaying air quality data and cannot help users determine the safety and health of outdoor activities such as jogging or cycling.
- AirAware introduces an AI-based recommendation system that analyzes the Air Quality Index (AQI), weather conditions, and pollution levels to tailor recommendations for safer outdoor activity locations and times for different users.

2.2.3 Lack of Clear and Accessible Pollution Education

- Although air pollution information is readily available online and free of charge, it is often presented in obscure technical jargon, making it difficult for many users to understand and discouraging them from actively seeking this information.
- AirAware simplifies complex environmental data through concrete educational content, clear visualizations, and interactive explanations, helping users better understand air pollution and its health impacts, whether actively or passively.

2.2.4 Limited Localized Information and Community Awareness

- Many existing global platforms only provide global air quality data, but lack a specific focus on the local situation in Malaysia or at the community level.

- AirAware will provide localized air pollution information, including regional news, pollution sources, and upcoming environmental events in the nearby area (such as Arbor Day), helping users understand the local environmental situation.

2.2.5 Lack of Interactive AI Assistance

- Most existing platforms only provide static information and lack interactive support for users.
- AirAware integrates a conversational AI assistant, allowing users to ask questions about air pollution, health risks, and protective measures at any time and receive instant and easy-to-understand answers

3.0 Methodology

3.1 Non-Technical Track

A.) Design thinking process:

Emphathize:

Understanding how users interact with the platform related to the topic of air pollution by exploring the existing website and mobile applications along with online research to have an understanding of the users' expectation of using the related website in the community discussion platform. Therefore, the research revealed that providing the guidance with the support of an AI assistant is more helpful for users compared to only showing the AQI values at different places. These findings help identify current limitations that can be innovated to enhance the user experience.

Define:

Based on insights explored during the empathize phase, we ascertained that a myriad of websites mainly emphasize on displaying AQI values that monitor for diverse locations on the maps show the scientific indicators that are difficult for users to understand without the support of AI

assistants. Accordingly, some of the users would not understand how AQI levels have the risk to affect their body health. Henceforth, there is a need for a solution that not only displays the AQI monitoring on the map but also offers AI-driven guidance to help users have a personalized health advisor which allows them to know the action needed to take in the environment with polluted air.

Ideate:

Several key features were proposed after the defined stage, namely real-time air quality display on interactive map, AI Chatbox**,** Personal Health Advisor, Smart AI assistant provide functions such as Smart Route Suggestion and Smart Driving Health which has a chance to integrate into electricity vehicles to support a healthier lifestyle. Additionally, Air Insight sections provide environmental information and keep updating on upcoming environmental events which could enhance the public awareness to alleviate the issue of air pollution.

Prototype:

After listing out all the features based on the problems required to address, we conducted a sketching and wireframing process to design the layout of the website in order to translate our idea into a functional interface to visualize how users would interact with the AirAware platform. Accordingly, we used Google AI Studio to develop our prototype via vibe coding for demonstrating the functionality of the AirAware platform.

Testing:

The prototype was reviewed to obtain the feedback from users and do the enhancement of the website in terms of designing, ways of displaying the data related to the air pollution for the sake of enabling users to have a better user experience.

B.) User research:

We conducted user research through reviewing the websites and applications coupled with reading the comments at community discussion at Reddit platform to figure out how the data related to the air pollution is presented on the platform for users to interact with it.

As a result, we noticed that a myriad of websites mainly emphasize on displaying AQI values that monitor for diverse locations on the maps and show the scientific indicators that are difficult for users to understand without the support of AI assistants. Accordingly, some of the users would not understand how AQI levels have the risk to affect their body health. Henceforth, there is a need for a solution that not only displays the AQI monitoring on the map but also offers AI guidance to help users have a personalized health advisor which allows them to know the action needed to take in the environment with polluted air.

C.) Flow creation:

1. The user opens the platform.
2. The user signs up the account and fills their health profile.
3. The user logs in the website after signing up.
4. The user views the AQI values on the AQI gauge for their current location.
5. The user explores the 24-hour and 7-day air quality forecast to understand pollution trends.
6. The user opens the interactive map to view the AQI level of different places they are looking for.
7. The user clicks the notification icon to view the personalized report card generated by AI Health Advisor in AI Chatbox.
8. The user asks the question they want to ask at the AI Chatbot to obtain the answer related to the topic of air pollution.
9. The user opens Air Insight to read the latest global and local news, upcoming environmental events, basic knowledge of air pollution and the tips to sustain a healthier lifestyle during air pollution.

10. The user opens Smart AI to choose the activity they tend to carry out and receive the Smart Route Suggestion.
11. The user who drives the electricity car is able to receive the driving health insight at Smart AI section.

4.0 Results

4.1 Outcome of the AirAware Platform

The creation of the AirAware prototype is a great example of how artificial intelligence can enhance our understanding of air pollution and help us to make healthier lifestyle choices. This platform brings together a variety of features that include real-time air quality monitoring, an AI chatbot, tailored health recommendations and smart route suggestions.

The interface for the prototype was created using Google AI Studio while employing vibe coding to create a simulated website experience. Users can easily access the air quality data through visual tools such as the AQI gauge and interactive complex environmental information into a format that is simple for users to understand.

However, with the help of AI Chatbot and Personal Health Advisor, users can ask questions about the air pollution and get personalized advice based on their health needs. This feature very effectively connects raw pollution data with actionable steps that users could take to safeguards their health.

Additionally, the Smart AI Assistant also offers the best route suggestions for outdoor activities such as jogging or cycling and it will be tailored to the current air quality conditions. The Smart Driving Health feature also could showcase a conceptual link with electric vehicle monitoring pollution exposure while driving for the users.

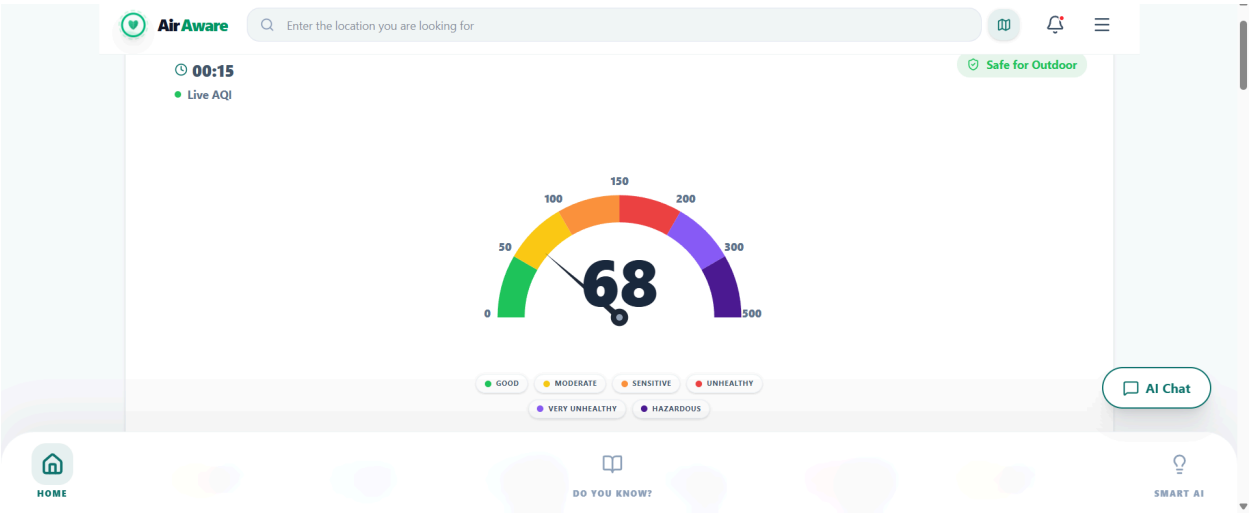
In conclusion, this project has successfully achieved its goals of raising awareness about air pollution and encouraging safer outdoor activities and delivering personalised health guidance to the users.

5.0 Prototype Showcase

This section is to introduce the prototype of the AirAware platform by showcasing its main user interface and essential features. The screenshots below illustrate how the users can engage with the platform and navigate various functionalities by also including air quality monitoring, AI-driven health recommendations and personalized dashboard insight.

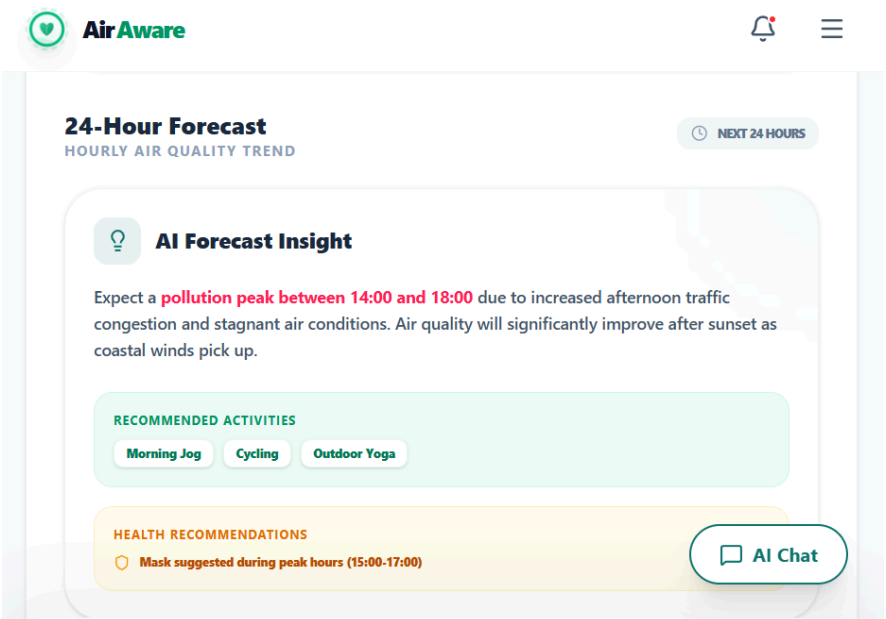
5.1 Output / UI Screenshots

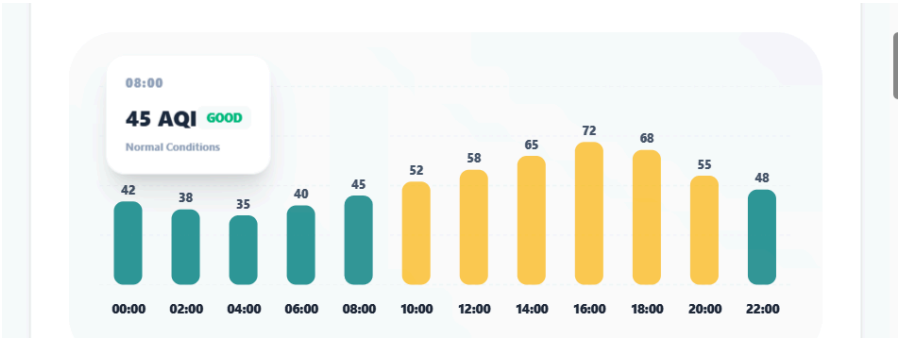
1. **Home Page of AirAware**



Captions: The Home Page provides an overview of the AirAware platform. Users can navigate to different system features and view general information about air quality monitoring and access the platform's main services.

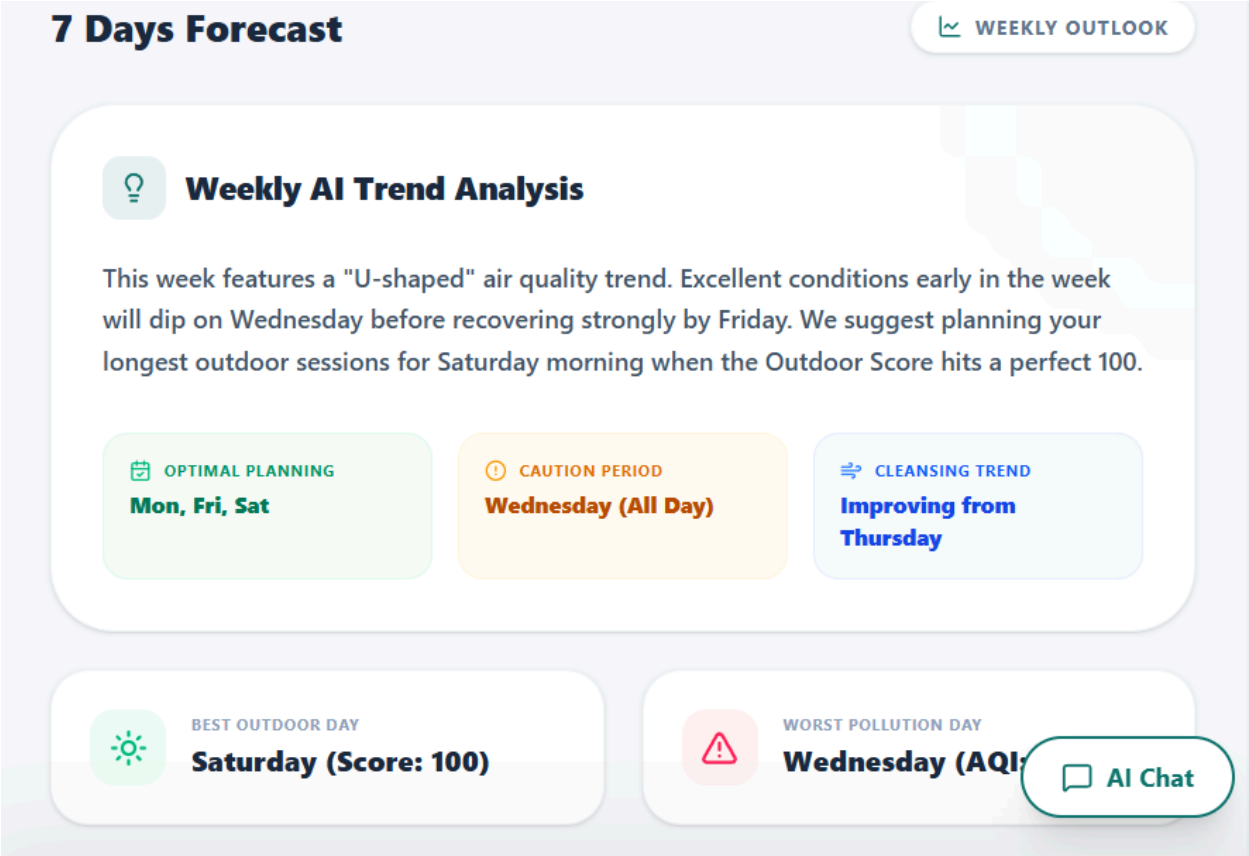
2. Air Quality Index (AQI) Page





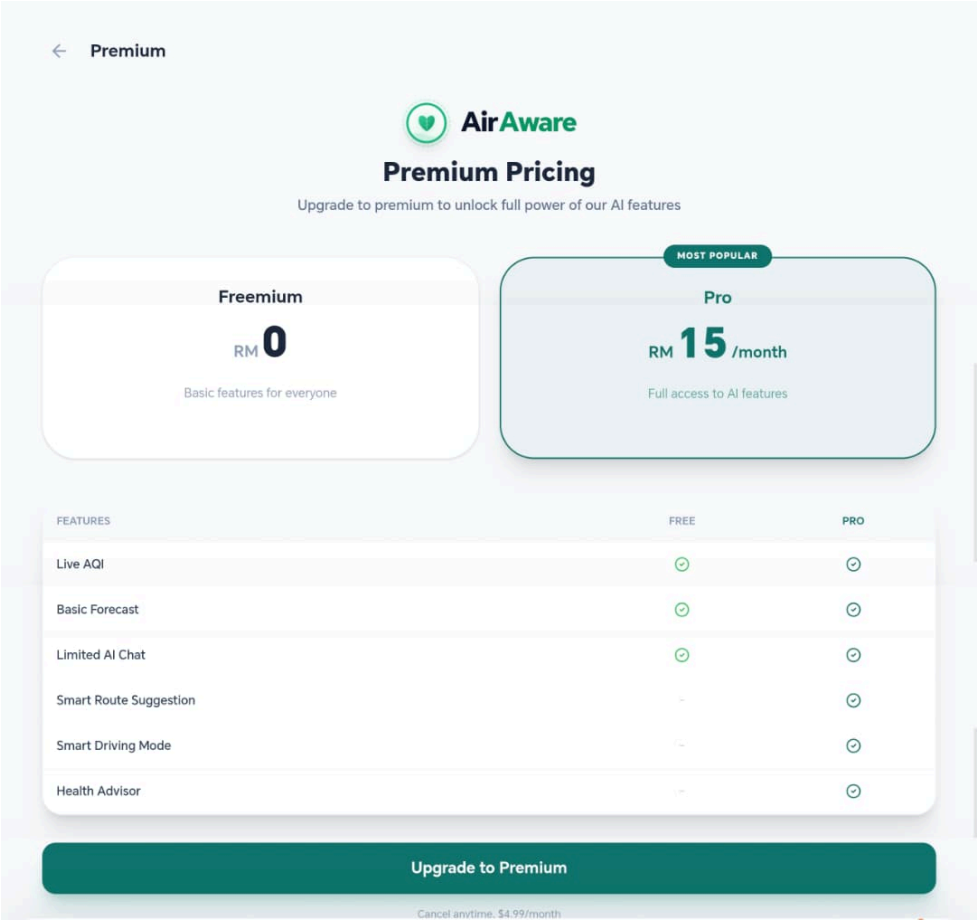
Caption: The AQI page is to display air quality information and environmental data, it allows users to monitor pollution levels and the understanding of the current air quality condition by the visual indicators and the data presentation.

3. **AI Health Advisor Interface**



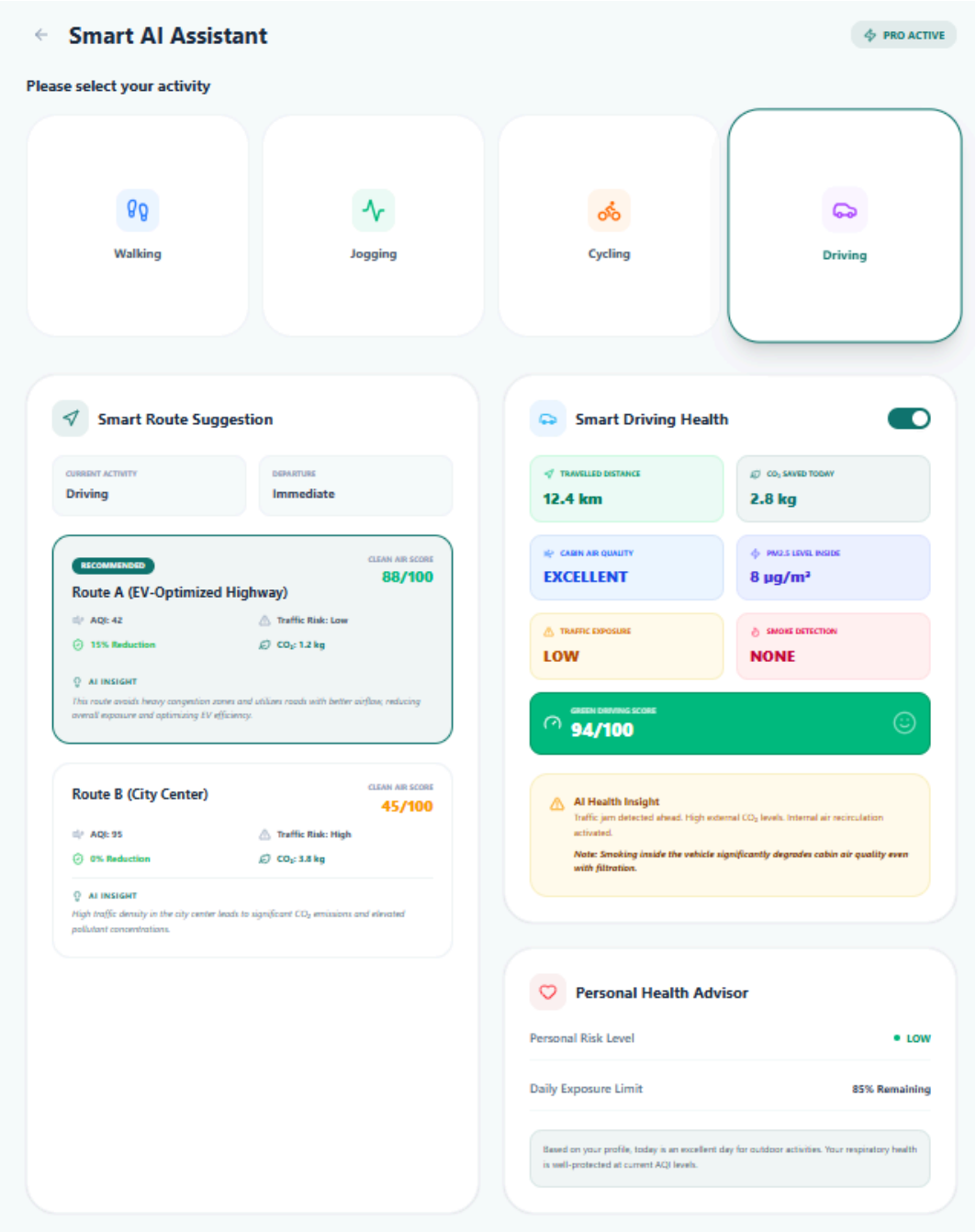
Captions: The AI Health Advisor provides users with health-related recommendations based on air quality conditions. Users can interact with the AI assistant to receive suggestions on how to reduce health risks during poor air quality situations.

4. Premium Features Page



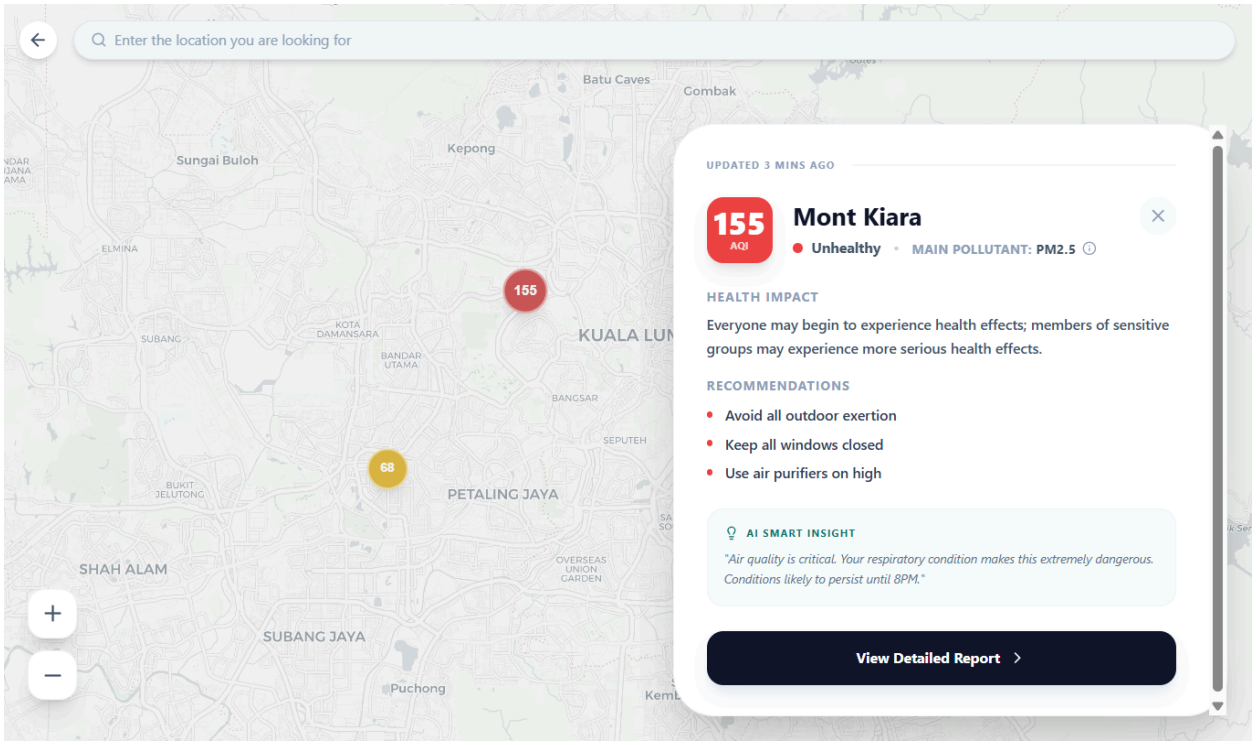
Captions: The premium page is to represent the subscription options available on the AirAware platform. Users can upgrade their account to unlock additional features and provide better services.

5. User Dashboard



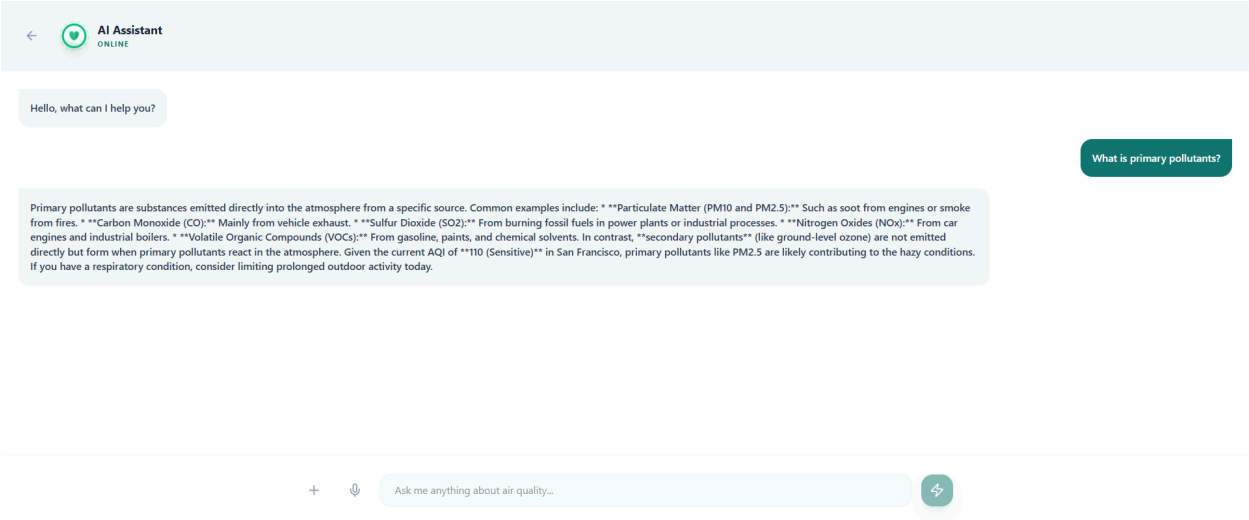
Captions: The user Dashboard provides a personalized overview of the system information by allowing the user to track air quality updates and access the different features in a centralized interface.

6. **Real-time air quality levels display on interactive map**



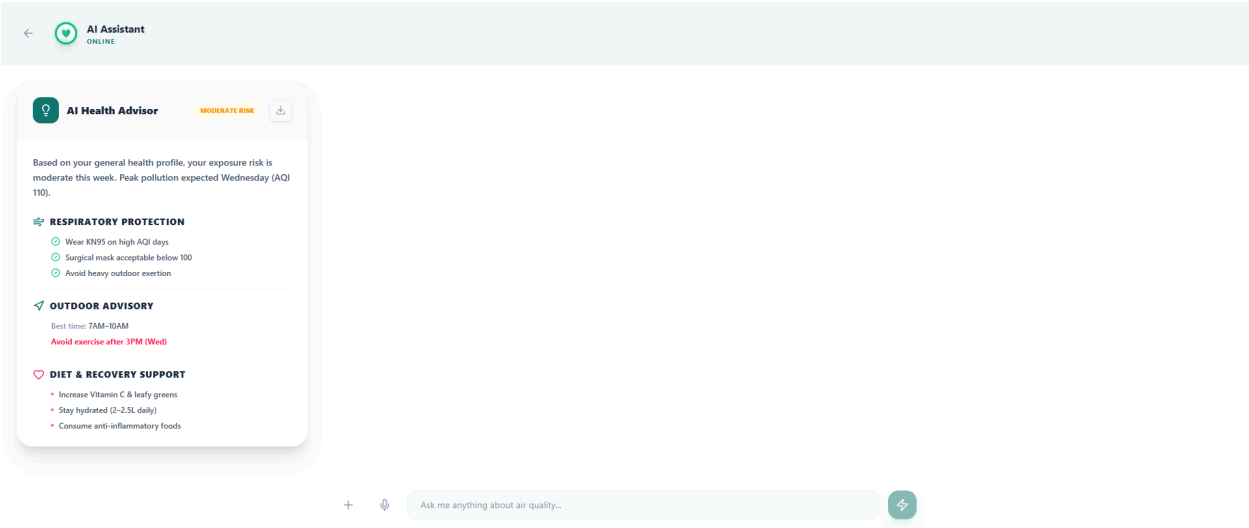
Captions: This map is to show the AI-driven and it also simply explains the health impact and it can recommend the action could be taken in a polluted environment.

7. AI ChatBox



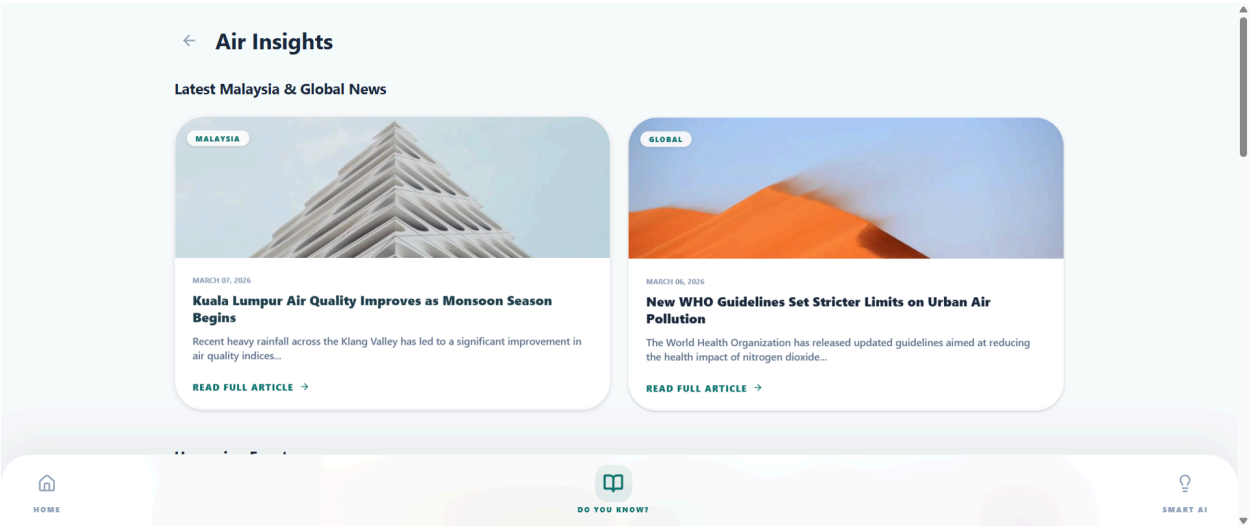
Captions: Users can ask questions about the topic of air pollution to enhance their understanding such as air quality status, health risks, pollution sources and recommended precautions.

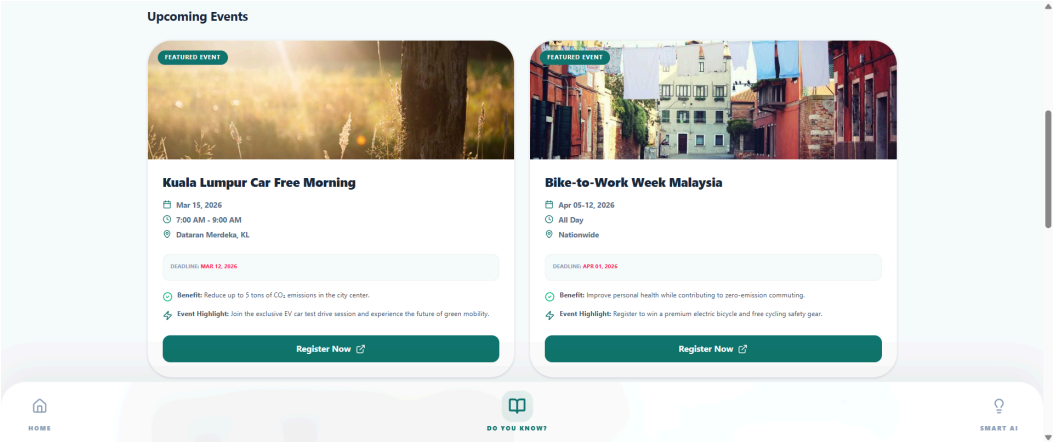
8. Personal Health Advisor

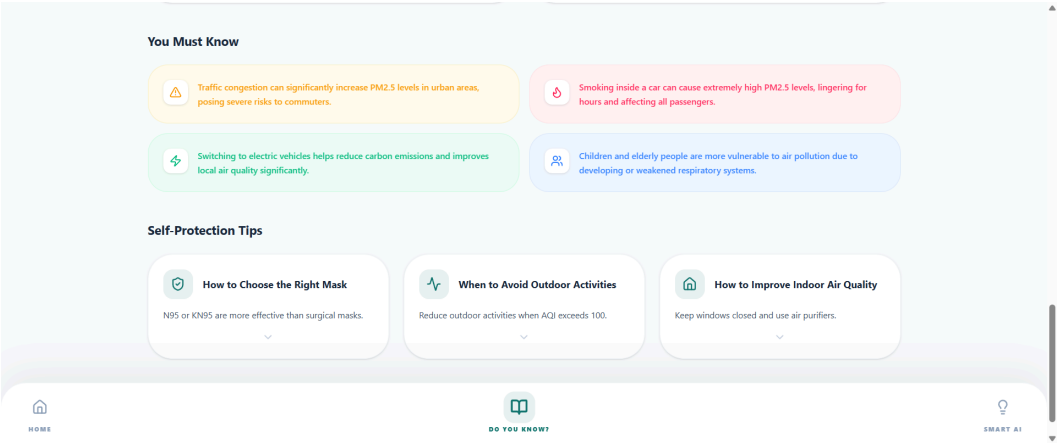


Captions: AI assistants can give several suggestions such as respiratory protection tips, outdoor advisory and ways to have a proper diet during air pollution based on the information from health profiles filled by users.

9. Air Insights







Captions: Provide latest global and local news about the air pollution and environmental issues.

6.0 AI Acknowledgement (Mandatory)

- ChatGPT used to check the logic ideas that need to apply in AirAware, rephrasing the sentence in the report, exploring the possible details that can add in the features when facing bottleneck and suggest the improvement for the layout of the website design.
- Google AI Studio is a tool to develop the prototype through vibe coding which is capable of helping us demonstrate our key features through presenting the clickable interface on the website.
- GitHub used as a sharing platform to store and manage the prototype files.

7.0 Conclusion

AirAware is an AI-powered platform to enhance the public awareness among urban residents, outdoor activities users and drivers using electric car by integrating the real-time AQI monitoring, AI Chatbot, AI personalized health advisor, AI recommendations based on real time situation and the environmental insights which are offering the knowledge related to air pollution in supporting users to have a sustainable healthy lifestyle. In addition, Smart AI providing the features like Smart Route Suggestion and Smart Driving Health has the potential to reduce the exposure to the polluted environment.

In the future, several enhancements could be considered to modify the AirAware platform. For example, collaboration with the electric vehicle manufacturer (EV) has an opportunity to integrate the Smart Driving Health Features into the vehicle system. Furthermore, we can also design a feature that is selling air pollution protection products to assist the related companies to promote their products.